

Assignment:2

Numpy Questions:

1. Create a NumPy array of shape (5, 5) filled with random integers between 1 and 20. Replace all the elements in the third column with 1.
2. Create a NumPy array of shape (6, 6) with values from 1 to 36. Extract the sub-array consisting of the 3rd to 5th rows and 2nd to 4th columns.
3. Create two NumPy arrays of shape (3, 4) filled with random integers. Perform element-wise addition, subtraction, multiplication, and division.
4. Given the NumPy array:

`A = [12, 45, 7, 89, 23, 56, 89, 34]`

- a) Find the **index of the maximum value** in the array.
 - b) Find the **index of the minimum value** in the array.
 - c) If there are **multiple maximum values**, identify all their indices.
5. Create a NumPy array of shape (5, 5) filled with random integers. Compute the mean, median, standard deviation, and variance of the array.

Pandas Questions:

6. Create a Pandas DataFrame with columns 'A', 'B', 'C' and index 'X', 'Y', 'Z'. Fill the DataFrame with random integers and access the element at row 'Y' and column 'B'.
7. Create a Pandas DataFrame with 3 columns and 5 rows filled with random integers. Add a new column that is the product of the first two columns.
8. Create a Pandas DataFrame with 3 columns and 5 rows filled with random integers. Introduce some NaN values. Fill the NaN values with the mean of the respective columns.
9. Create a Pandas Series with 5 random text strings. Extract the first three characters of each string.
10. Create two Pandas DataFrames with different columns. Concatenate the DataFrames along the rows and along the columns.
11. Perform the following questions **on the basis of the given CSV file (student_data.csv)** using the Pandas library.
 - i. Import the Pandas library and load the data from the file **student_data.csv** into a Pandas DataFrame. Display the **first five records** of the dataset.

- ii. From the DataFrame created using `student_data.csv`, display only the **Name**, **Class**, and **Math_Marks** columns.
- iii. Using the data from `student_data.csv`, retrieve and display the details of students whose **Attendance_Percentage is greater than 85**.
- iv. Calculate and display the **average Math_Marks** of all students present in the file `student_data.csv`.
- v. Sort the data from `student_data.csv` in **descending order of Science_Marks** and display the **top five student records** after sorting.

Matplotlib Questions:

12. Plot Customization

Create a line plot for the following data:

- `x = [0, 1, 2, 3, 4]`
- `y = [5, 10, 15, 20, 25]`

Customize your plot with:

- A red dashed line.
- Square markers with size 8.
- A title: "Customized Line Plot".
- Axis labels: "X Values" and "Y Values".
- Custom tick marks for the x-axis: `[0, 2, 4]`

13. Bar Chart

Create a bar chart representing the following data:

- Categories: `["Apples", "Bananas", "Cherries", "Dates"]`.
- Values: `[15, 25, 10, 20]`.

Add:

- A title: "Fruit Sales".
- An x-axis label: "Fruit".
- A y-axis label: "Quantity Sold".
- Different colors for each bar.

14. Scatter Plot

Using the following data:

- `x = [1, 2, 3, 4, 5, 6]`
- `y = [2, 4, 1, 8, 7, 3]`

Create a scatter plot with:

- Circular markers.
- Marker sizes corresponding to values in y multiplied by 20.
- A legend labeled "Data Points".
- A grid in the background

15. Saving a High-Resolution Plot

Generate a quadratic curve plot ($y = x^2$) for x ranging from -10 to 10.

Customize the plot with:

- A green solid line.
- Labels for the x and y axes: "X Values" and " $Y = X^2$ ".
- Save the figure as "quadratic_curve.png" with a resolution of 300 dpi.